

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)

Activity Tool : MCQ Test (Low)

Assessment Tool Weightage: 30%

Dr.G.Ayyappan

QUESTIONS Total Marks: 10			
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	1
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	1
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	1
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	1
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	1
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	1

7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	1
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	1

9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	1
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	1

Code of Conduct:

I **KESHAVARAM L/192511253** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action. **Signature of the student:**

L. Keshavam

RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect

Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

QUESTION 1

Question

Which of the following is true about 1NF?

Options

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic
- (d) Eliminates transitive dependencies

Correct Answer

- (c) All attributes must be atomic**

Explanation

First Normal Form (1NF) requires every attribute value to be atomic, meaning each cell contains a single indivisible value. Multi-valued attributes and repeating groups are not allowed. Partial dependencies are addressed by 2NF, while transitive dependencies are addressed by 3NF.

Key Concept

1NF → Atomic attribute values

QUESTION 2

Question

A relation is in 2NF if it is in 1NF and:

Options

- (a) Has no transitive dependencies

- (b) Has no partial dependencies on primary key
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Correct Answer

(b) Has no partial dependencies on primary key

Explanation

2NF requires the relation to already be in 1NF and requires every non-prime attribute to be fully functionally dependent on the whole candidate key. Therefore, partial dependency on only part of a composite key must be eliminated. Transitive dependencies are handled by 3NF.

Key Concept

2NF $\rightarrow$ 1NF + no partial dependency

QUESTION 3

Question

Which normal form deals with multi-valued dependencies?

Options

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

Correct Answer

(d) 4NF

Explanation

Fourth Normal Form (4NF) specifically addresses non-trivial multi-valued dependencies (MVDs). A relation in 4NF should not contain a non-trivial MVD $X \twoheadrightarrow Y$ unless X is a superkey.

Key Concept

4NF $\rightarrow$ Multi-Valued Dependencies

5. QUESTION 4

Question

BCNF is a stricter version of:

Options

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 4NF

Correct Answer

(c) 3NF

Explanation

Boyce-Codd Normal Form (BCNF) is stricter than 3NF. In BCNF, for every non-trivial functional dependency $X \rightarrow Y$, X must be a superkey. Some relations can satisfy 3NF but still violate BCNF.

Key Concept

BCNF $\rightarrow$ Stronger form of 3NF

6. QUESTION 5

Question

Functional dependency $X \rightarrow Y$ means:

Options

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

Correct Answer

- (b) X determines Y

Explanation

A functional dependency $X \rightarrow Y$ means that whenever two tuples have the same value for X, they must also have the same value for Y. Thus, X functionally determines Y. The reverse dependency $Y \rightarrow X$ is not automatically implied.

Key Concept

$X \rightarrow Y$ means X determines Y

7. QUESTION 6

Question

Which of the following anomaly is NOT associated with unnormalized relations?

Options

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly

Correct Answer

- (d) Retrieval anomaly

Explanation

Insertion, deletion and update anomalies are the classic anomalies caused by poor or unnormalized relational design. Retrieval anomaly is not one of the standard three anomalies identified in normalization theory.

Key Concept

8. QUESTION 7

Question

A superkey is:

Options

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key

Correct Answer

- (b) Any set of attributes that uniquely identifies tuples**

Explanation

A superkey is any set of one or more attributes that uniquely identifies every tuple in a relation. It does not have to be minimal. A candidate key is a minimal superkey, which makes option (a) the definition of a candidate key rather than a superkey.

Key Concept

Superkey = Unique identification; Candidate key = Minimal superkey

9. QUESTION 8

Question

If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

Options

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A
- (d) Independently determined

Correct Answer

- (c) Transitively dependent on A**

Explanation

Because $A \rightarrow B$ and $B \rightarrow C$, A determines C through the intermediate attribute B. Therefore, C is transitively dependent on A. This is the pattern $A \rightarrow B \rightarrow C$ that 3NF is designed to address.

Key Concept

$A \rightarrow B$ and $B \rightarrow C \Rightarrow A \rightarrow C$ transitively

10. QUESTION 9

Question

5NF is also known as:

Options

- (a) Domain-Key Normal Form
- (b) Boyce-Codd Normal Form
- (c) Project-Join Normal Form
- (d) Element Normal Form

Correct Answer

(c) Project-Join Normal Form

Explanation

Fifth Normal Form (5NF) is also called Project-Join Normal Form (PJ/NF). It deals with non-trivial join dependencies and ensures that a relation cannot be further decomposed without losing necessary information, except where the decomposition is implied by candidate keys.

Key Concept

5NF = Project-Join Normal Form (PJ/NF)

11. QUESTION 10

Question

Lossless-join decomposition ensures:

Options

- (a) No data is lost during decomposition
- (b) All FDs are preserved
- (c) The relation is in BCNF
- (d) No null values exist

Correct Answer

(a) No data is lost during decomposition

Explanation

A decomposition is lossless when joining the decomposed relations reconstructs the original relation without generating spurious tuples or losing valid information. Dependency preservation and BCNF are separate properties; a lossless decomposition does not automatically guarantee either of them.

Key Concept

Lossless join $\rightarrow$ Original relation can be reconstructed correctly